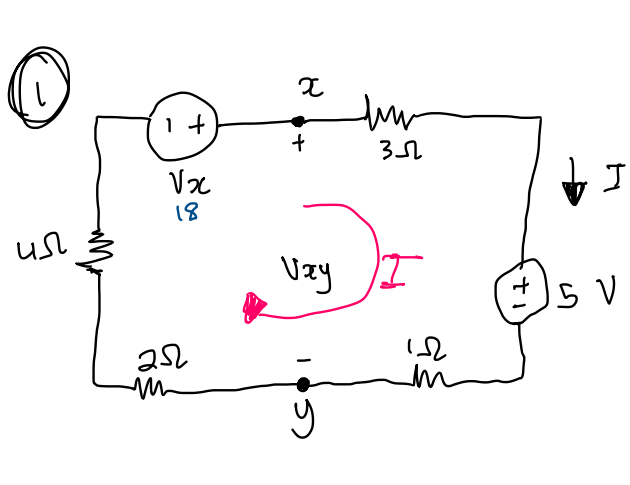
Assessment 2

Sunday, 06 August 2023 21:14

①

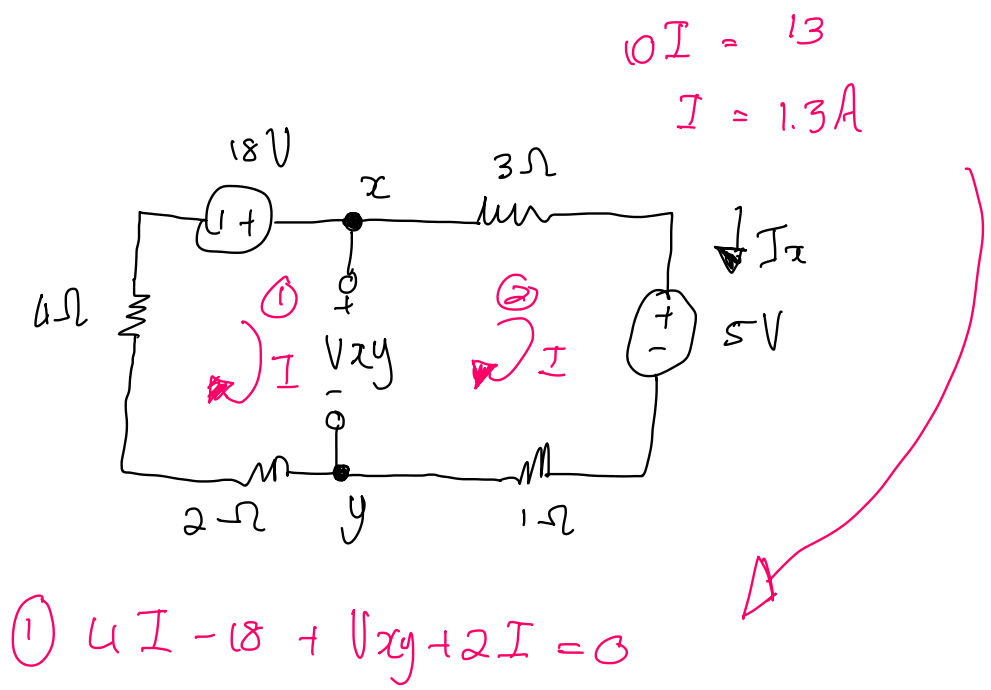

x
3Ω
Ix
4Ω
Vx
18
Vxy
I
5 V
2Ω
1Ω
y

$4I - 18 + 3I + 5 + 1I + 2I = 0$

$10I - 13 = 0$

$10I = 13$

$I = 1.3A$


18 V
x
3Ω
Ix
4Ω
①
②
I
Vxy
I
5V
2Ω
y
1Ω

① $4I - 18 + V_{xy} + 2I = 0$

$4(1.3) - 18 + V_{xy} + 2(1.3) = 0$

$V_{xy} = 10.2V$

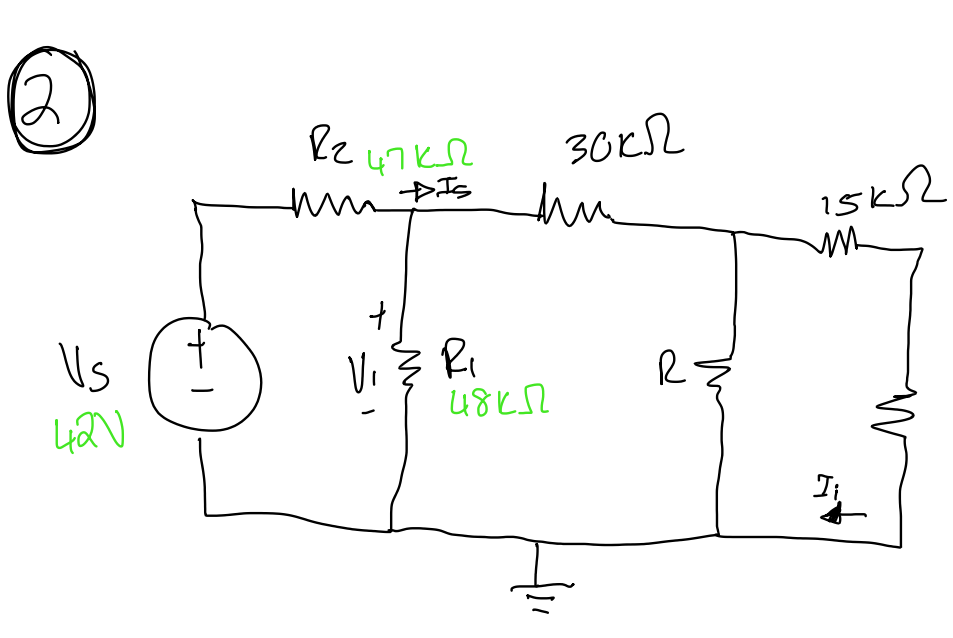
② $-V_{xy} + 3I + 5 + 1I = 0$

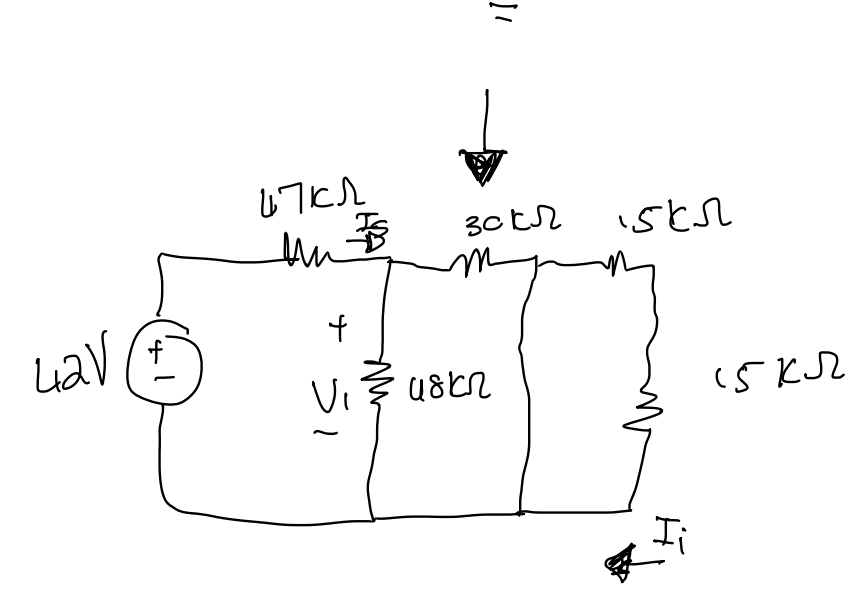
$-V_{xy} + 4(1.3) + 5 = 0$

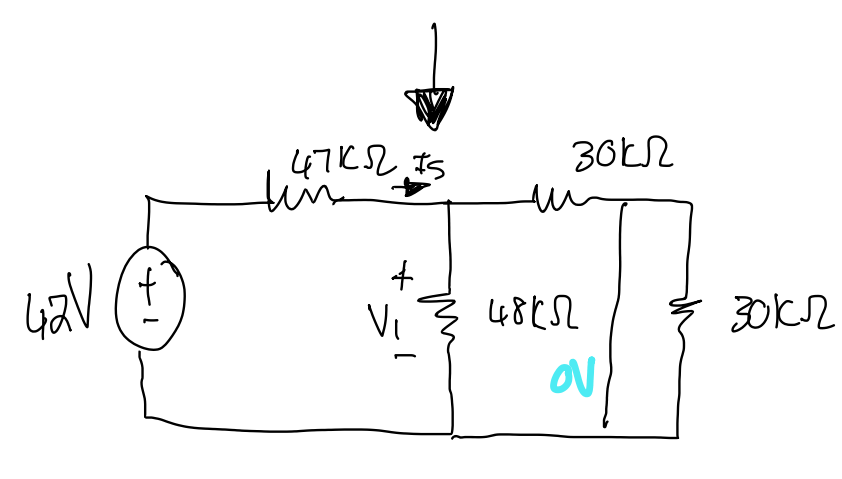
$-V_{xy} = -10.2V$

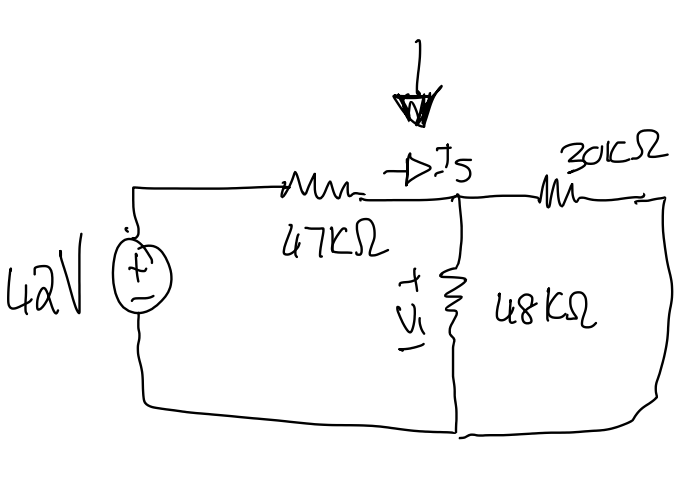
$V_{xy} = 10.2V$

②


R2 47kΩ
Is
30kΩ
15kΩ
Vs
42V
V1
R1
48kΩ
R
15kΩ
I1

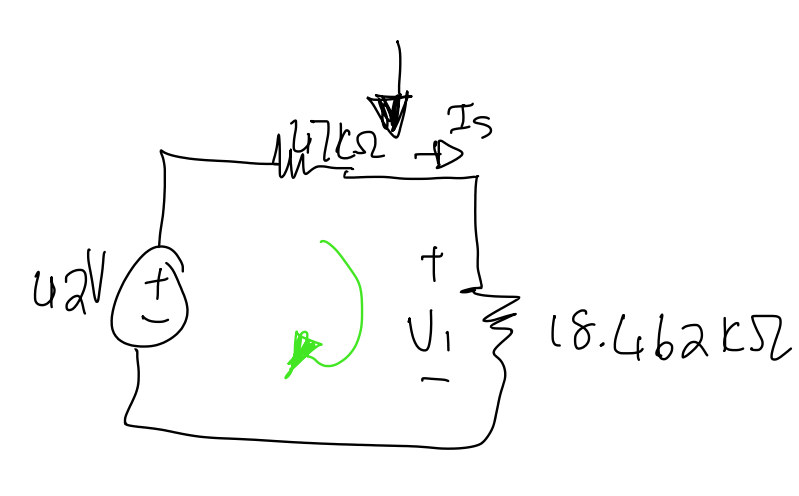

47kΩ
Is
30kΩ
15kΩ
42V
V1
48kΩ
15kΩ
I1


47kΩ
Is
30kΩ
42V
V1
48kΩ
0V
30kΩ


Is
30kΩ
42V
47kΩ
V1
48kΩ

$R_p = \left(\frac{1}{48} + \frac{1}{30}\right)^{-1}$

$= 18.462$


47kΩ
Is
42V
V1
18.462kΩ

$-42 + 47000 I_s + 18462 I_s = 0$

$65462 I_s = 42$

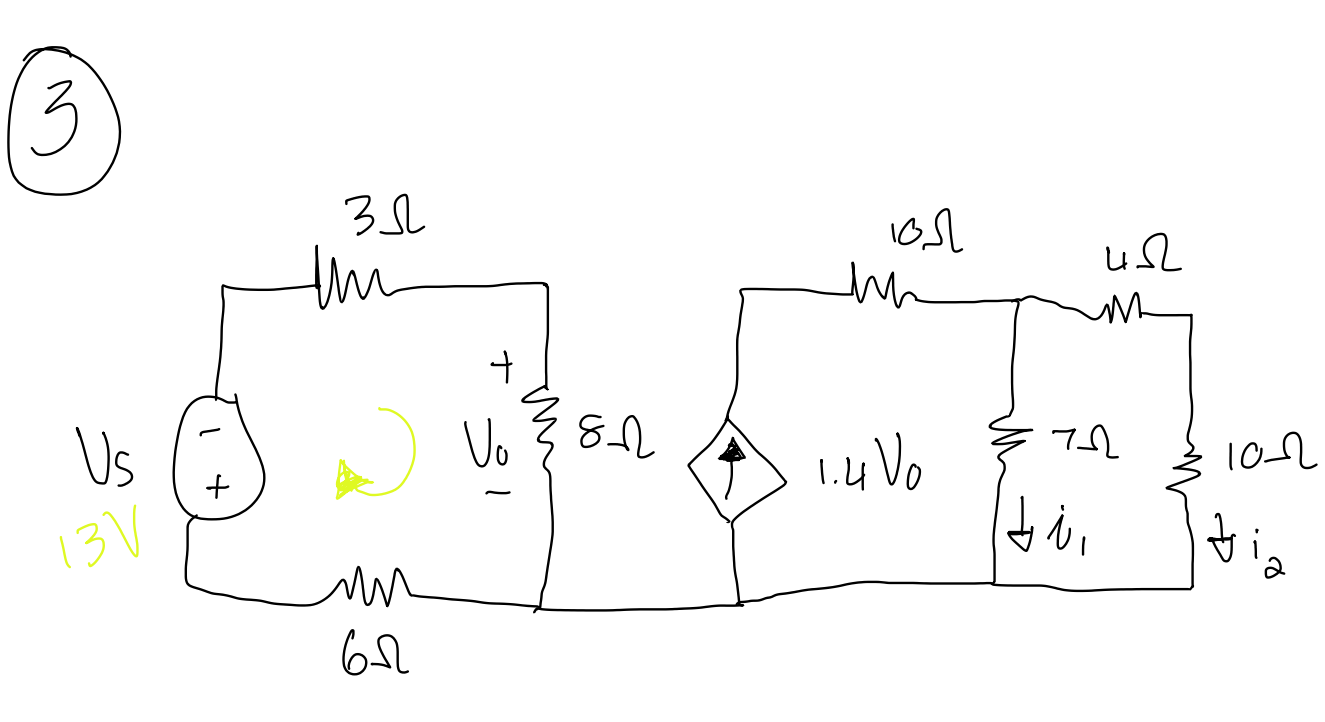
$I_s = 0.000641598$

$\therefore V_1 = \pm [IR]$

$= +(0.000641598)(18462)$

$= 11.845 V$

③


3Ω
Vs
13V
Vo
8Ω
6Ω
1.4Vo
10Ω
4Ω
7Ω
i1
10Ω
i2

$13 + 3I + 8I + 6I = 0$

$17I = -13$

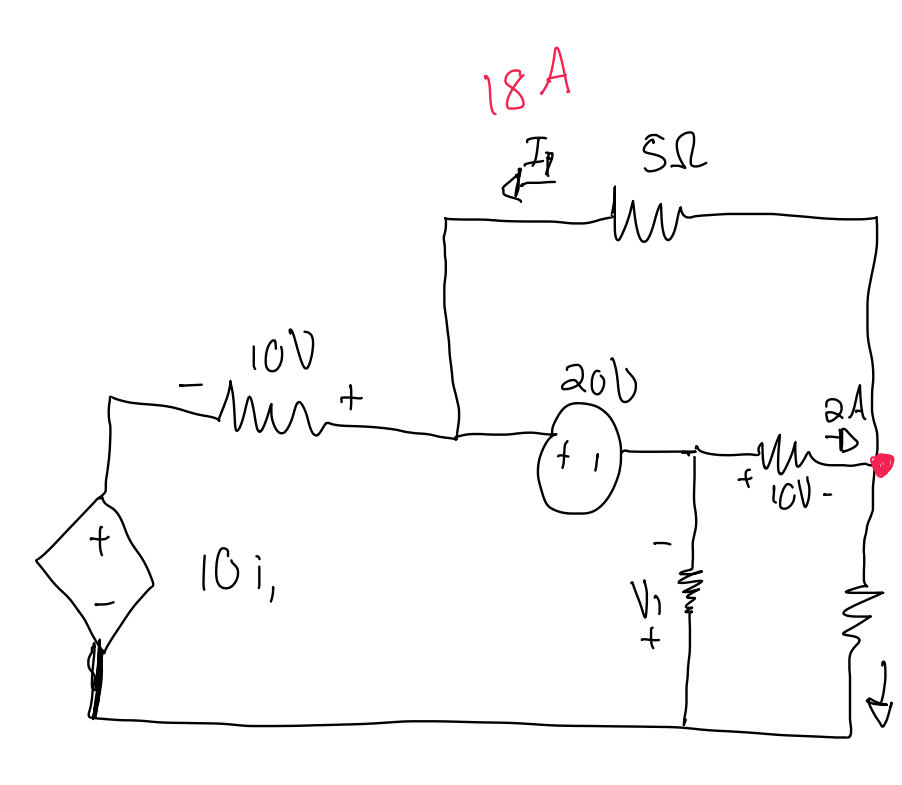
$I = -0.76471 A$

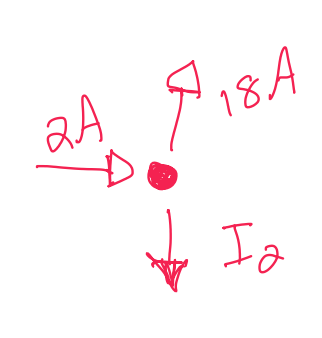
$\therefore V_0 = \pm [IR]$

$= +(-0.76471)(8)$

$= -6.1176 V$

④


18A
I1
5Ω
10V
20V
2A
10V
10i1
V1
I2



In = out

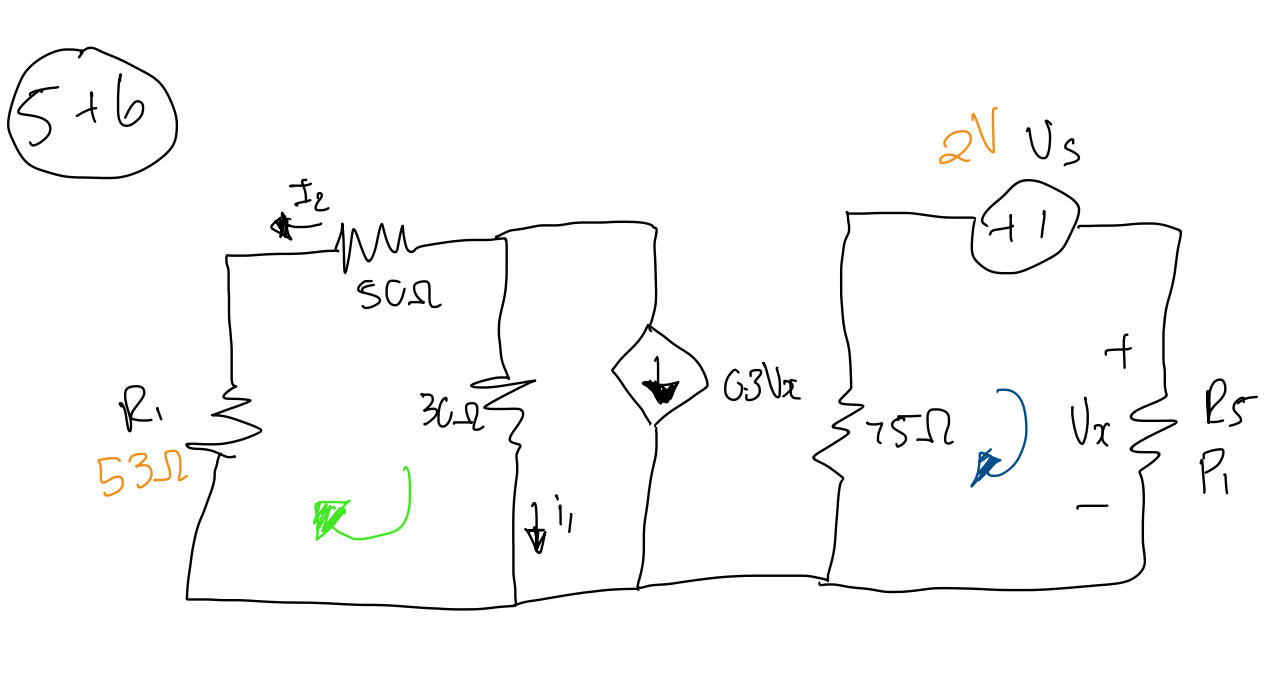
$2 = 18 + I_2$

$-16A = I_2$

$\frac{11}{20} V_1 - \frac{1}{10} V_2 = \frac{10}{5}$

$-\frac{1}{10} V_1 + \frac{19}{20} V_2 = 2$

⑤+⑥


I2
50Ω
R1
53Ω
30Ω
i1
0.3Vx
2V Vs
75Ω
Vx
R5 = 43Ω
P1

⑤ $75I + 2 + 43I = 0$

$118 I = -2$

$I = -0.017 A$

$V_x = \pm [IR]$

$= +(-0.017)(43)$

$= -0.73 V$

$\therefore P_1 = \pm [VI]$

$= +(-0.73)(-0.017)$

$= 0.012353 W$

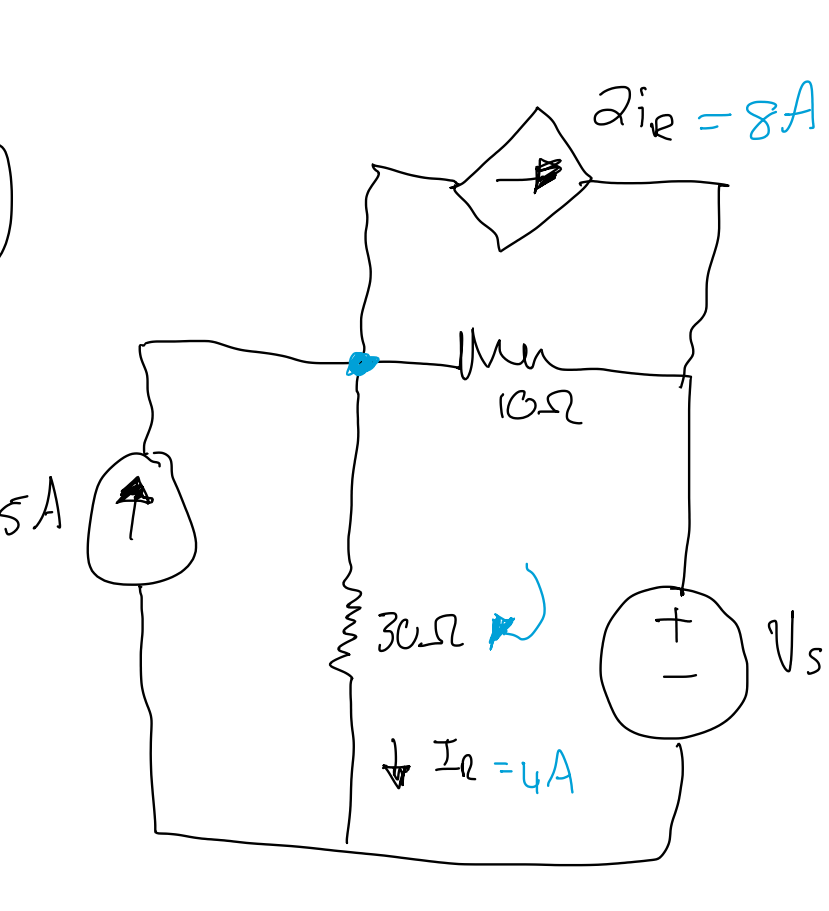
$= 12.353 mW$ Ⓕ

⑥ $\left(\frac{50+53}{53+50+30}\right)(-0.3V_x)$

$= \left(\text{" "}\right)(0.21)$

$= 0.16933 A$

⑦


2ie = 8A
5A
10Ω
30Ω
Ie = 4A
Vs

① 5 8 4 Ip

③ $I_p = I$

$I = -I_e$

② In = out

$5 = 8 + 4 + I_p$

$I_p = -7A$

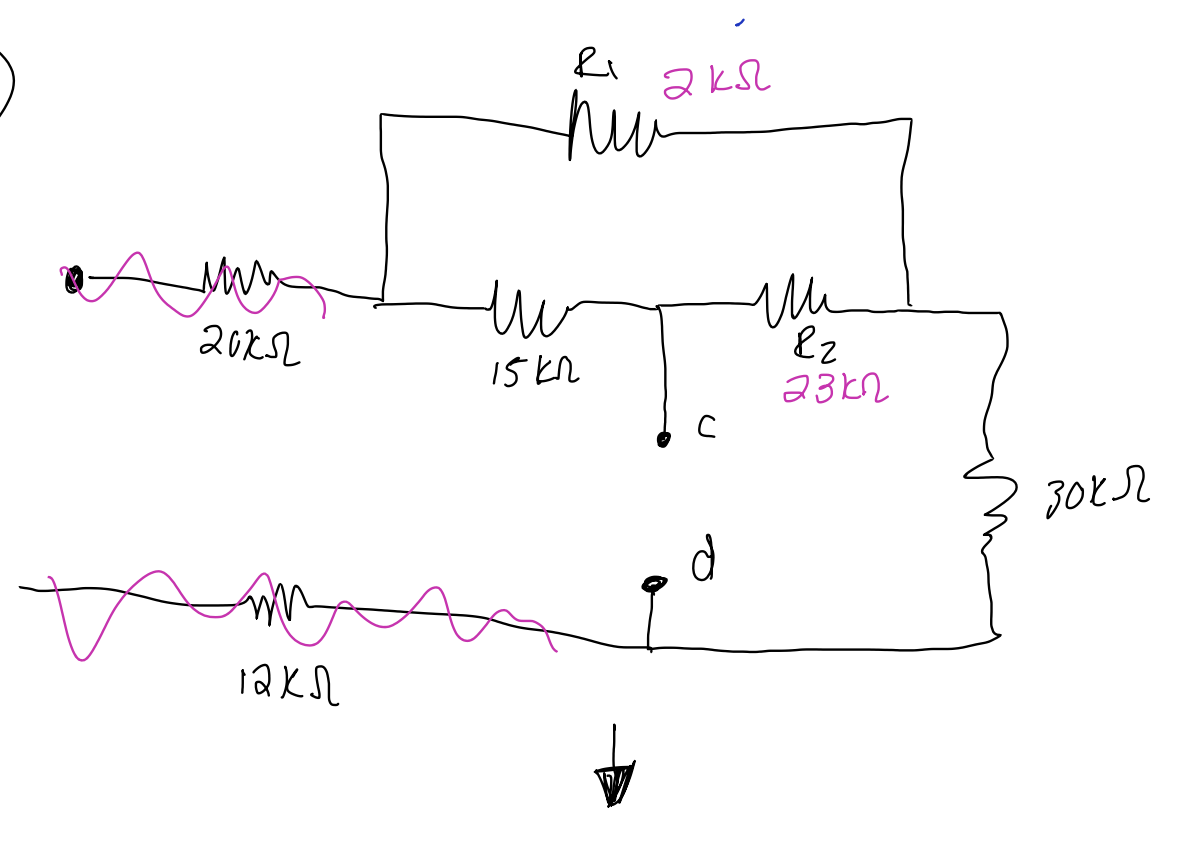
④ KVL

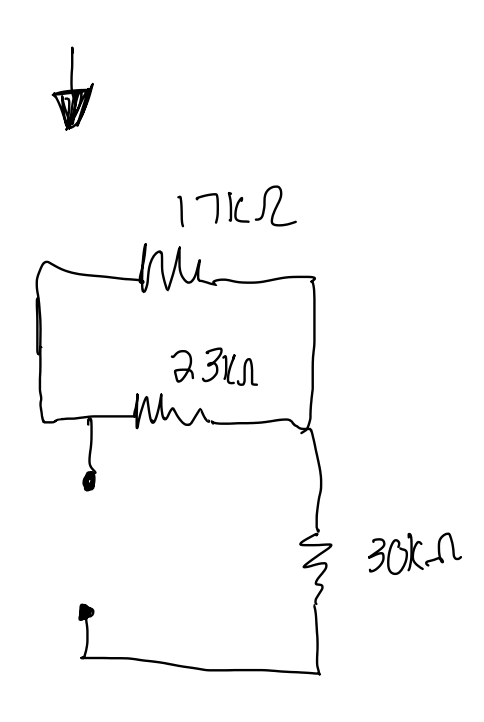
$10 I_p + V_s + 30 I_e = 0$

$10(-7) + V_s + 30(-4) = 0$

$V_s = 190V$

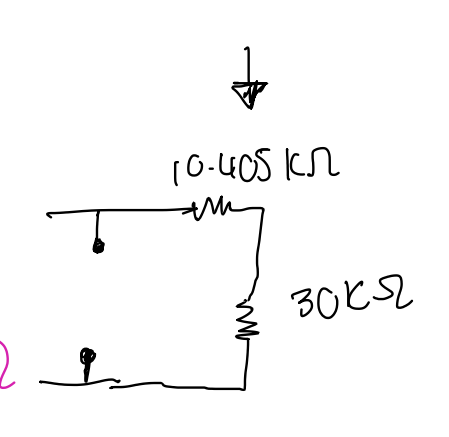
⑧


R1 2kΩ
20kΩ
15kΩ
R2
23kΩ
c
d
30kΩ
12kΩ


17kΩ
23kΩ
30kΩ

$R_p = \left(\frac{17 \times 23}{17+23}\right)$

$= 9.775 k\Omega$

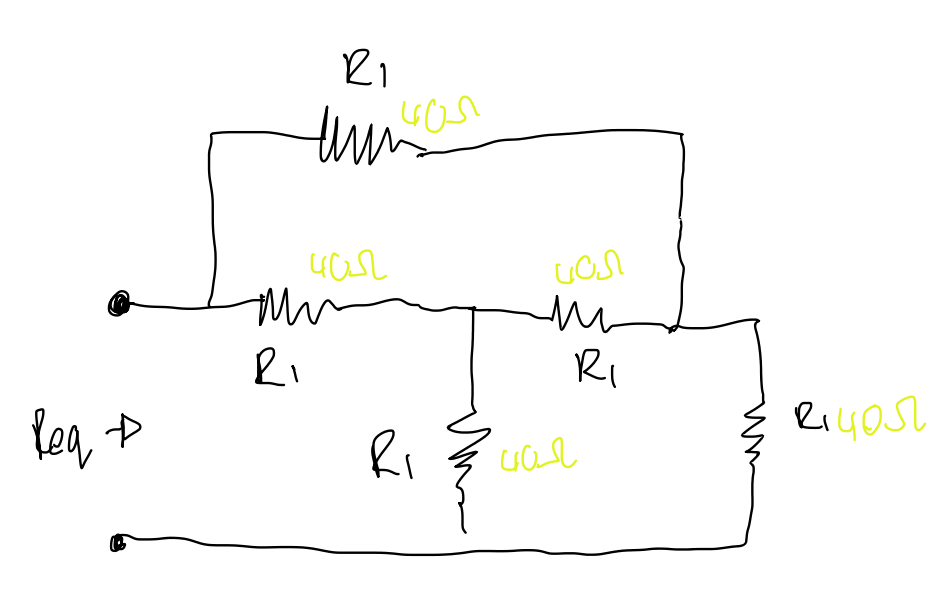

10.405kΩ
30kΩ

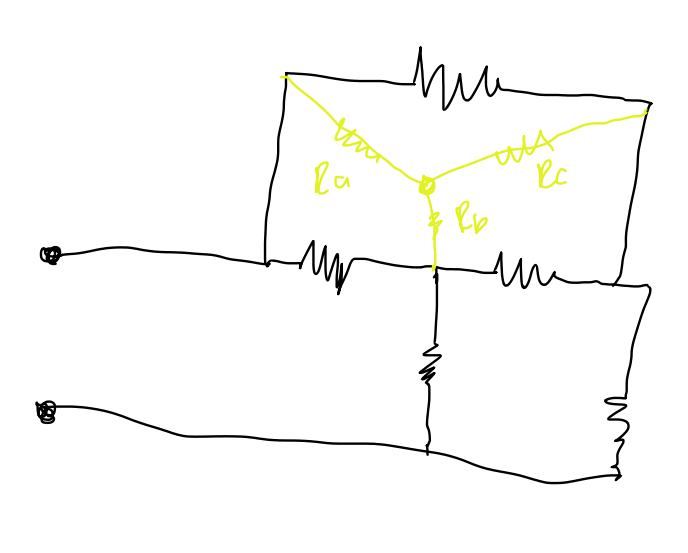
$R_{eq} = R_p + 30$

$= 9.775 + 30$

$= 39.775 k\Omega$

⑨

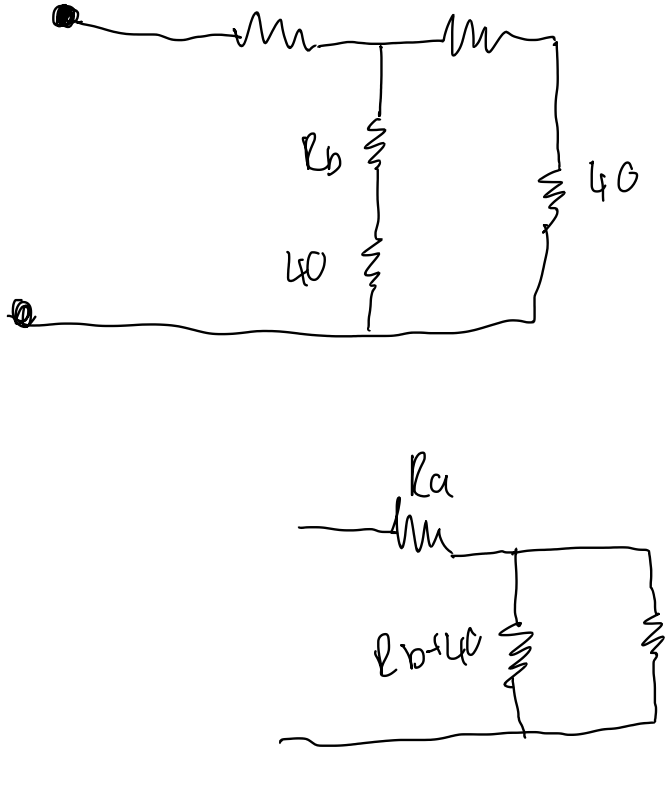

R1
40Ω
40Ω
40Ω
R1
R1
Req
R1
40Ω
R1 40Ω

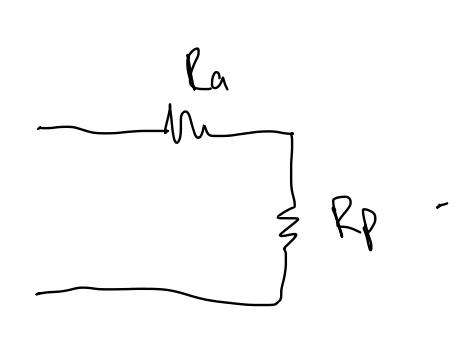


$R_a = \frac{40 \times 40}{40+40+40} = 13.33\Omega$

$R_b = \frac{40 \times 40}{40+40+40} = 13.33\Omega$

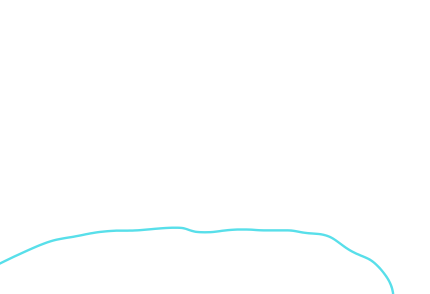
$R_c = \frac{40 \times 40}{40+40+40} = 13.33\Omega$


Ra
Rc
Rb
40
40


Ra
Rb+40
Rc+40

$R_p = \left(\frac{1}{R_c + 40} + \frac{1}{R_b + 40}\right)^{-1}$

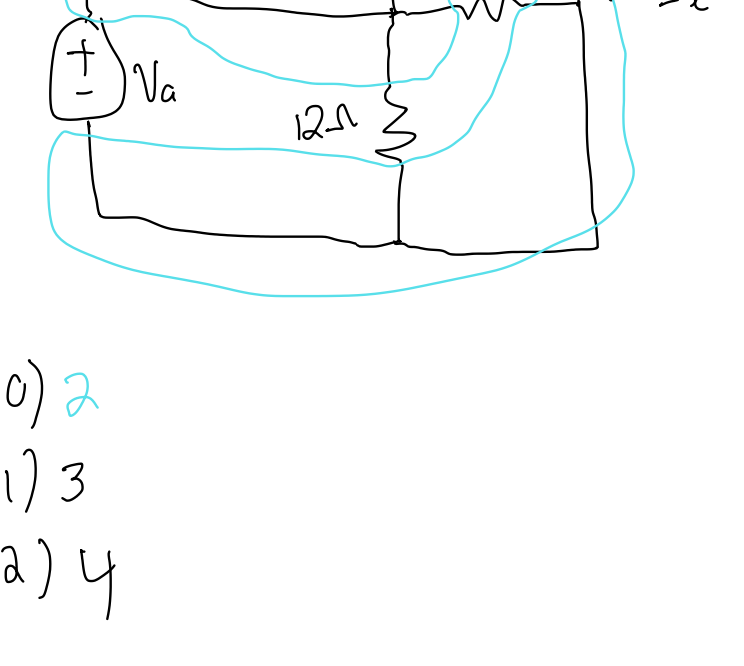
$= 26.667 \Omega$



$\therefore R_{eq} = R_a + R_p$

$= 40 \Omega$

(10-12)

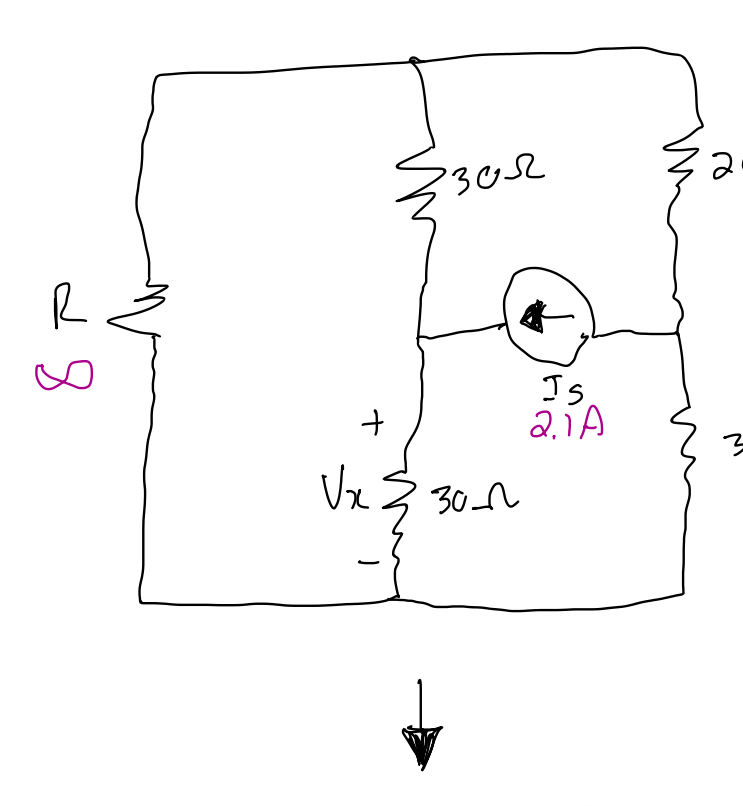

10Ω
Vx
24Ω
4Ω
Ix
Va
12Ω

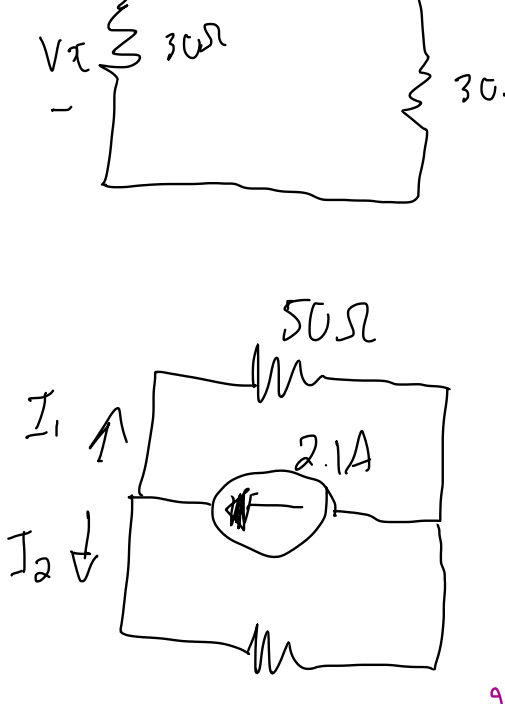
10) 2

11) 3

12) 4

⑬


30Ω
20Ω
R
∞
Is
2.1A
Vx
30Ω
30Ω

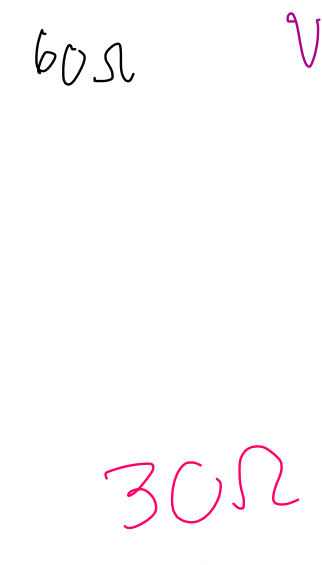

30Ω
Is = 2.1A
20Ω
Vx
30Ω
30Ω

$R_s = 30 + 20$

$= 50$

$R_s = 30 + 30$

$= 60$


50Ω
I1
I2
2.1A
60Ω

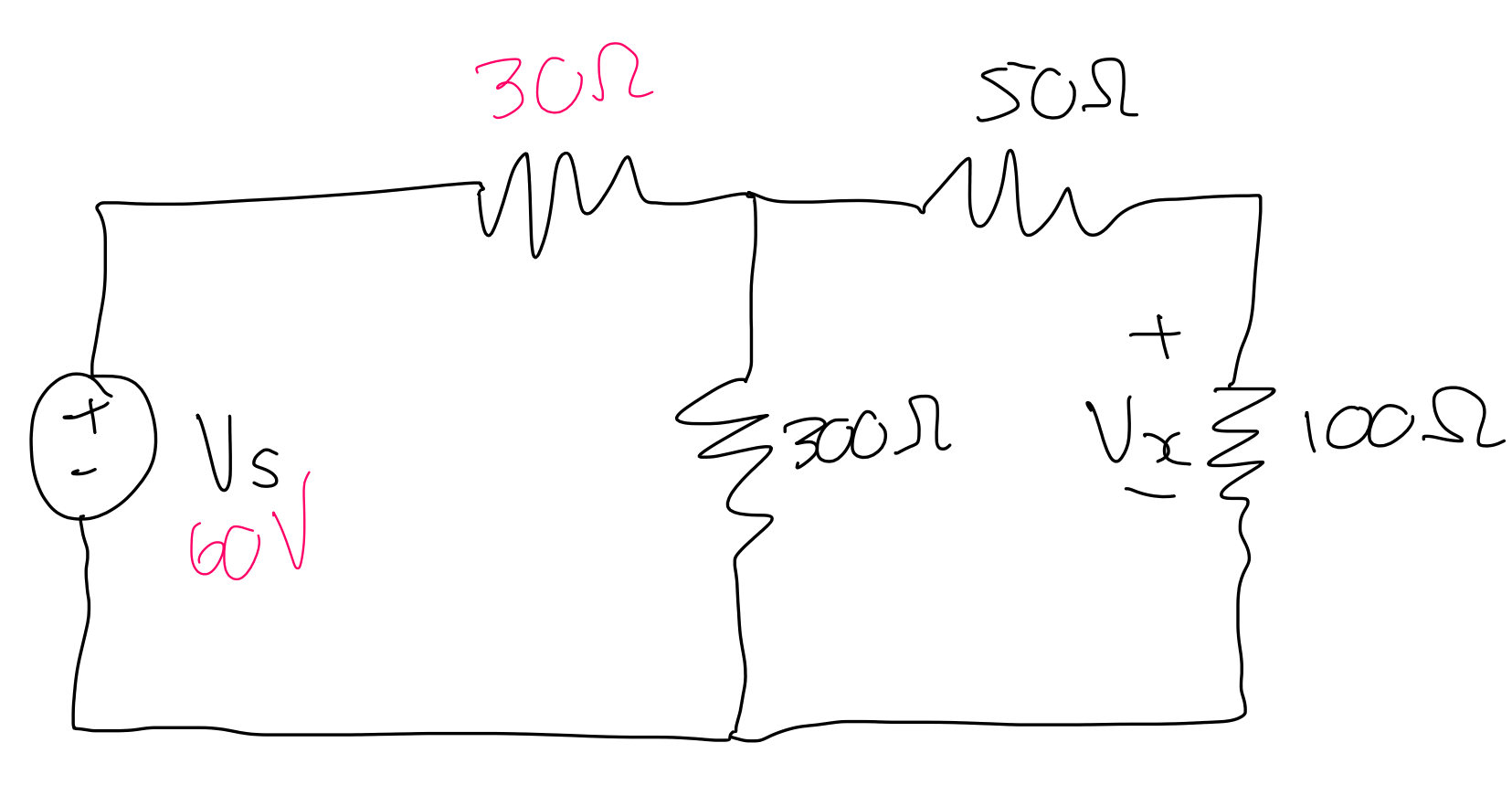
$I_2 = \left(\frac{50}{50+60}\right) \times 2.1$

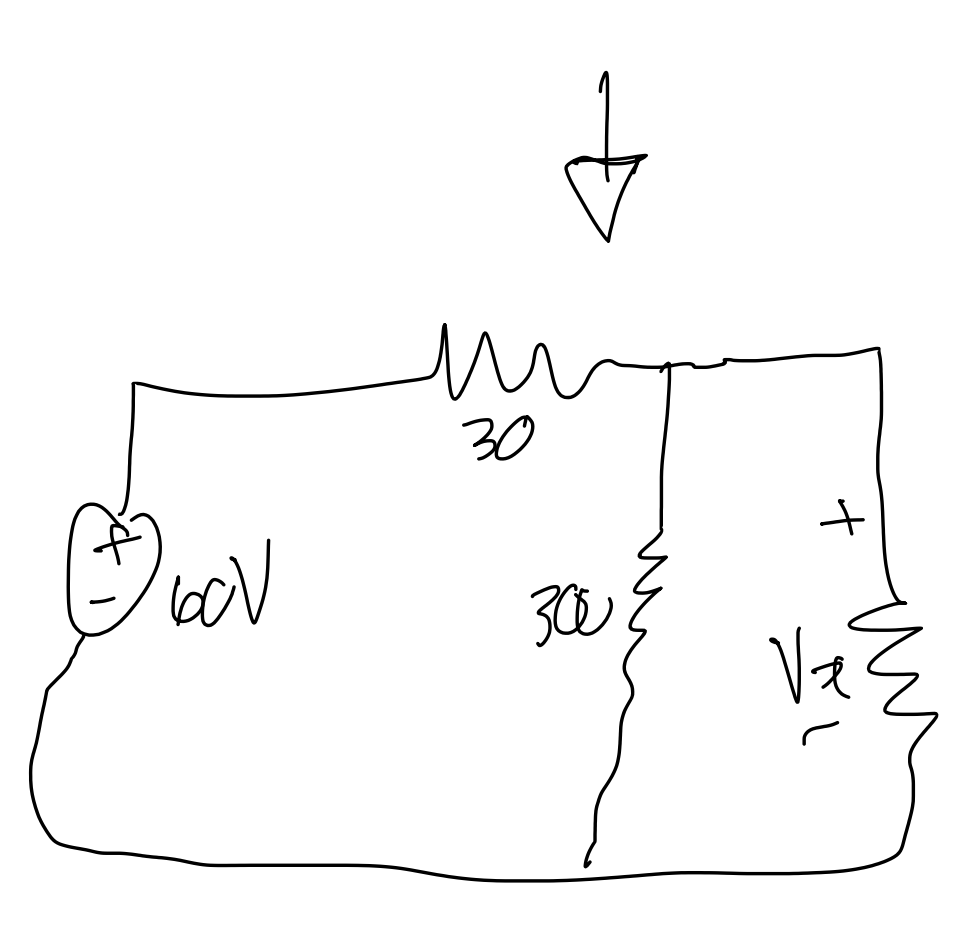
$= 0.954545 A$

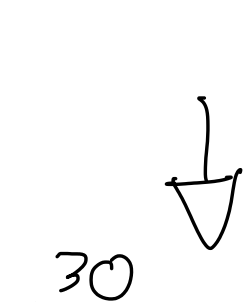
$V_x = \pm [IR]$

$= +(0.9545)(30)$

$= 28.636 V$


30Ω
50Ω
Vs
60V
300Ω
Vx
100Ω


30
60V
300
Vx
150Ω


30
60V
Vx
100Ω

$R_p = \left(\frac{1}{300} + \frac{1}{150}\right)^{-1}$

$= 100\Omega$

$-60 + 30I + 100I = 0$

$130I = 60$

$I = 0.4615 A$

$V_x = \pm IR$

$= +(0.4615)(100)$

$= 46.154 V$